

Jennifer Crodelle, PhD

Assistant Professor of Mathematics

Middlebury College

✉ jcrodelles@middlebury.edu 🌐 sites.middlebury.edu/jcrodelles

Research Interests

- **Neuronal Networks:** Understanding the dynamics of neuronal networks during development and diseased states such as epilepsy.
- **Pain Processing:** Uncovering mechanisms underlying pain processing in the spinal cord using mathematical modeling.

Academic Appointments

- 2020–Present** Assistant Professor of Mathematics, Middlebury College, Middlebury, VT
- 2017–2020** NSF Mathematical Sciences Postdoctoral Research Fellow (MSPRF), Courant Institute of Mathematical Sciences, New York University

Education

- Aug 2017** **Ph.D. in Mathematics**, Rensselaer Polytechnic Institute
- May 2012** **B.S. in Applied Mathematics**, Marist College

Publications

(* indicates undergraduate student co-authors)

- [1] A.G. Ginsberg, F.S. Lempka, B. Duan, V. Booth, **J. Crodelle**. Mechanisms for dysregulation of excitatory-inhibitory balance underlying allodynia in dorsal horn neural subcircuits. *PLoS Computational Biology*, 21(1): e1012234 (2025).
- [2] X. Du, **J. Crodelle**, V. J. Barranca, S. Li, Y. Shi, et al. Biophysical modeling and experimental analysis of the dynamics of *C. elegans* body-wall muscle cells. *PLoS Computational Biology*, 21(1): e1012318 (2025).
- [3] **J. Crodelle**, C. Vanty*, and V. Booth. Modeling homeostatic and circadian modulation of human pain sensitivity. *Frontiers in Neuroscience*, 17: 1166203 (2023).
- [4] J. M. Epstein, E. Hatna, **J. Crodelle**. Triple contagion: a two-fears epidemic model. *Journal of the Royal Society Interface*, 18(181): 20210186 (2021).
- [5] **J. Crodelle** and D. W. McLaughlin. Modeling the role of gap junctions between excitatory neurons in the developing visual cortex. *PLoS Computational Biology*, 17(7): e1007915 (2021).

- [6] **J. Crodelle**, C. Vallejo, M. Schmidtchen, C. Topaz, and M.R. D’Orsogna. Impacts of California Proposition 47 on crime trends in Santa Monica, CA. *PLoS ONE*, 16(5): e0251199 (2021).
- [7] **J. Crodelle** and P. Maia. A Computational model for pain processing in the dorsal horn following axonal damage to receptor fibers. *Brain Sciences*, 11(4): 505 (2021).
- [8] **J. Crodelle**, D. Zhou, G. Kovacic, and D. Cai. A computational model of electrotonic coupling between pyramidal cells in the cortex. *Journal of Computational Neuroscience*, 48(4): 387–407 (2020).
- [9] Z.Q. Tian, **J. Crodelle**, and D. Zhou. A Combined Offline-Online Algorithm for Hodgkin-Huxley Neuronal Networks. *Journal of Scientific Computing*, 84(1): 10 (2020).
- [10] **J. Crodelle**, K.A. Newhall, P.B. Pyzza, and G. Kovacic. Coarse-grained descriptions of oscillations in neuronal network models. *Communications in Mathematical Sciences*, 17(5): 1437–1458 (2019).
- [11] **J. Crodelle**, M. Hagenauer, S. Piltz, and V. Booth. Modeling the daily rhythm of human pain processing in the dorsal horn. *PLoS Computational Biology*, 15(7): e1007106 (2019).
- [12] **J. Crodelle**, D. Zhou, G. Kovacic, and D. Cai. A role for electrotonic coupling between cortical pyramidal cells. *Frontiers in Computational Neuroscience*, 13: 33 (2019).
- [13] Z.Q. Xu, **J. Crodelle**, D. Zhou, and D. Cai. Maximum entropy principle analysis in network systems with short-time recordings. *Physical Review E*, 99: 022409 (2019).
- [14] **J. Crodelle**, M. Hagenauer, S. Piltz, and V. Booth. A neural circuit model for pain processing in the spinal cord. *Proceedings of A Research Collaboration Workshop for Women in Mathematical Biology*, Springer (2016).
- [15] M. Hagenauer, **J. Crodelle**, S. Piltz, N. Toporikova, P. Ferguson, and V. Booth. The Modulation of Pain by Circadian and Sleep-Dependent Processes: A Review of the Experimental Evidence. *Proceedings of A Research Collaboration Workshop for Women in Mathematical Biology*, Springer (2016).

Journal Articles in Progress

- **J. Crodelle** and W. Dai. Activity-dependent effect of cholinergic waves on LGN to V1 synapse formation.
- A. Byrne and **J. Crodelle**. Mathematical approach to understanding gap junctions and seizure-induced activity.

Grant Funding

2025	NIH R15 HEAL Initiative, \$375,000
2023–2025	Vermont Biomedical Research Network Project Award, \$150,000

2022–2023	Vermont Biomedical Research Network Pilot Award, \$25,000
2017–2020	National Science Foundation MSPRF, \$150,000 (DMS-1703761)

Teaching Experience

Middlebury College

Spring 2024	Numerical Analysis (MATH 0228)
S24, F21, S21, F20	Differential Equations (MATH 0226)
Spring 2022	Partial Differential Equations (MATH 0326)
Winter 2022	Data Science Across Disciplines (NSCI/MATH 1230)
F22, S22, F21	Calculus II (MATH 0122)
F22, S21	Mathematical Modeling (MATH 0315)
Fall 2020	Multivariable Calculus (MATH 0223)

Courant Institute, New York University

S20, F19, S19	Linear Algebra (MATH 140)
Fall 2018	Ordinary Differential Equations (MATH 262)

Invited Seminars

Dec 2024	Modeling the effect of spontaneous retinal waves on the development of receptive fields of neurons in primary visual cortex, <i>Applied Mathematics Seminar</i> , Courant Institute, NY
Oct 2024	A Modeling approach to studying the effect of retinal waves on the development of receptive fields of neurons in primary visual cortex, <i>Mathematical Biology Seminar</i> , Brandeis University (Virtual)
Oct 2024	Mathematical modeling of neuronal networks and their functional roles, <i>Mathematics Seminar</i> , Swarthmore College, Swarthmore, PA
Oct 2022	Exploring synchrony in the brain through mathematical modeling, <i>Pi Mu Epsilon Induction Ceremony</i> , Saint Michael's College, Colchester, VT
Apr 2022	Development of orientation preference in mice: a mathematical model, <i>RWTH Aachen University EDDY Seminar</i> (Virtual)
Oct 2020	A simple mathematical model of synapse formation in the developing visual cortex of mice, <i>Applied Mathematics Seminar</i> , UNC Chapel Hill, NC (Virtual)
Nov 2019	Modeling visual circuit development of mice through synaptic plasticity, <i>Simons Collaboration on the Global Brain Postdoc Meeting</i> , New York, NY

Oct 2019	Do mice and cats see eye-to-eye?, <i>Mathematics Colloquium</i> , Williams College, Williamstown, MA
Jun 2019	Introduction to computational neuroscience, <i>Undergraduate Summer Research Seminar</i> , Courant Institute, NY
Feb 2018	Circadian rhythmicity of pain sensitivity: A firing-rate model of dorsal horn circuitry, <i>Computational Biology Seminar</i> , Courant Institute, NY
Mar 2017	The role of gap junctions in synchronizing neuronal activity, <i>Oxford College of Emory University Colloquium</i> , Oxford, GA
Jan 2017	An Investigation into the role of gap junctions in synchronizing neuronal activity, <i>Marist College Seminar Series</i> , Poughkeepsie, NY

Selected Invited Conference Talks

Oct 2024	Modeling the activity-dependent effect of retinal waves on circuit formation in the visual cortex, <i>AMS Eastern Sectional Meeting</i> , Albany, NY
Jan 2023	Mathematical modeling in the brain: investigating the formation of network connections, <i>Joint Mathematics Meetings (JMM)</i> , Boston, MA
Jun 2021	Firing-rate models for analyzing spinal circuit motifs underlying chronic pain, <i>Society for Mathematical Biology (SMB) Annual Meeting</i> (Virtual)
Jul 2019	Modeling visual circuit development of mice through synaptic plasticity, <i>Society for Mathematical Biology (SMB) Annual Meeting</i> , Montreal, CAN
May 2019	Modeling gap junctions in the cortex, <i>SIAM Conference on Applications of Dynamical Systems</i> , Salt Lake City, UT
Aug 2018	Gap junctions between pyramidal cells in cortical neuronal networks, <i>SIAM Conference on The Life Sciences</i> , Minneapolis, MN
Jun 2017	Synchrony among synaptically and electrically connected neurons in the cortex, <i>Third International Conference on Mathematical Neuroscience</i> , Boulder, CO

Selected Contributed Conference Presentations

Oct 2024	Mathematical Model of stage II retinal waves and their effect on the development of receptive fields for neurons in primary visual cortex (poster), <i>Society for Neuroscience (SfN) Annual Meeting</i> , Chicago, IL
Jun 2024	A functional role for stage II retinal waves on the receptive field development of neurons in primary visual cortex (poster), <i>International Conference on Mathematical Neuroscience</i> , Dublin, Ireland
Nov 2022	Mathematical modeling approach to investigating inhibitory plasticity in the visual cortex (poster), <i>Society for Neuroscience (SfN) Annual Meeting</i> , San Diego, CA

Aug 2022	Plasticity among neurons in the visual cortex during development: a mathematical modeling approach (poster), <i>MathFest</i> , Philadelphia, PA
Oct 2021	Mathematical modeling of neuronal networks, <i>Fall Faculty Forum</i> , Middlebury, VT
Nov 2018	A mathematical model for the circadian rhythmicity of pain sensitivity in the dorsal horn (poster), <i>Society for Neuroscience (SfN) Annual Meeting</i> , San Diego, CA

Awards & Honors

2023–Present	Jeanne Epp Barksdale Junior Faculty Fellowship (\$6,000 yearly development fund)
May 2023	Perkins Award for excellence in teaching (\$5,000 development fund)
Fall 2021	Middlebury CTLR Pedagogy Enrichment Funds
2017–2020	NSF Mathematical Sciences Postdoctoral Fellowship (\$150,000, DMS-1703761)
May 2017	Joaquin B. Diaz Thesis Prize at Rensselaer Polytechnic Institute
Apr 2015	Student Paper Award at the 9th IMACS International Conference, Athens, GA

Mentoring Experience

Research Mentor / Advisor to:

- **Liza Platonov & Riley Hale** (Middlebury, Summer 2024): Undergraduate RAs working on modeling gap junction (GJ) gating dynamics.
- **Lila Kosowsky & Izzy Pentony** (Middlebury, Winter 2024): Undergraduate RAs focused on inhibitory plasticity.
- **Alex Ginsberg** (University of Michigan, 2021–2023): PhD student advised by Victoria Booth. Served as mentor and committee member; thesis defended July 2023.
- **Ai Hattori** (Middlebury, Summer 2022): Undergraduate RA modeling the effect of GJs in epileptic seizures.
- **Daniel Ellison** (Middlebury, Summer 2022): Undergraduate RA investigating two plasticity rules.
- **Carrie Vanty** (Middlebury, 2021–2022): Undergraduate RA who developed a mathematical model to explore the interaction of sleep and pain.
- **Bryan Currie** (Middlebury, Summer 2021): Undergraduate RA modeling synchronization properties of neurons coupled by a GJ.
- **Ben Elstner** (Middlebury, Summer 2021): Undergraduate RA characterizing inhibitory STDP.
- **Paulina Czarnecki** (University of Michigan, Summer 2019): Undergraduate RA at Courant program focused on modeling electrophysiological properties of a Merkel cell.

- **Taylor Meredith** (Courant, Summer 2018): Undergraduate student modeling Myasthenia Gravis and its treatment.

Workshops Attended

Oct 2024	OCCTIVE Workshop: teaching computing outside of computer science courses, Union College, Schenectady, NY
Jun 2021	SIMIODE Developer's Workshop: Differential Equations Model and Resource Creators, SIMIODE (Virtual)
Mar 2021	Mathematical and computational approaches to social justice, ICERM at Brown University, Providence, RI (Virtual)
Sep 2019	Statistical model fitting, NYU Center for Neural Science, New York, NY
Jul 2018	Crime in Santa Monica, AMS-MRC: Agent-Based Modeling in Biological and Social Systems, Whispering Pines, RI
Aug 2015	Understanding neuromechanical processes in locomotion, SAMSI: CCNS
Jun 2015	Sleep, circadian rhythms and pain, Workshop for Women in Mathematical Biology, NIMBioS, Knoxville, TN
May 2015	SIAM Workshop on Network Sciences, Salt Lake City, UT

Organizing Activities

Jan 2023	Co-organizer, Panel Session <i>Advancing your Career: Integrating Scholarship, Teaching, and Service</i> , Joint Mathematics Meetings, Boston, MA
May 2021	Co-organizer, Minisymposium <i>New dynamical systems frameworks for investigating neuronal network computations</i> , SIAM DS, Virtual
Jul 2019	Co-organizer, Minisymposium <i>Mathematical modeling of neuronal networks</i> , SMB Annual Meeting, Montreal, CAN
May 2019	Co-organizer, Minisymposium <i>Neuronal Computations in Brain Networks</i> , SIAM DS, Salt Lake City, UT
Aug 2018	Co-organizer, Minisymposium <i>Information Processing in Neuronal Networks</i> , SIAM LS, Minneapolis, MN
May 2017	Co-organizer, Minisymposium <i>Computational models of neuronal connectivity in the brain</i> , SIAM DS, Salt Lake City, UT

Travel Grants

May 2019	SIAM Early Career Travel Award
Jan 2019	AMS-MRC Travel Award

Aug 2018	SIAM Early Career Travel Award
Jul 2018	Institute of Natural Sciences (INS) Travel Award

Service & Outreach

Dec 2024	Judge for SCUDEM undergraduate mathematical modeling competition
2022–Present	Middlebury AWM Student Chapter Faculty Advisor
Aug 2022	Judge for MathFest undergraduate poster session
Jun 2022	Panelist for CTLR Graduate School Panel
May 2022	External committee member for thesis student Majd Hamdan (Physics, Middlebury)
Apr 2021	Judge for Middlebury Hackathon
Apr 2021	Intro to mathematical modeling talk for Middlebury Math Club
2020–2021	College Street Children’s Center Board Member
2020–2021	1000 Girls 1000 Futures Mentor
Apr 2019	Mathematical Modeling course at Courant Splash! (for high school students)
Mar 2018	Keynote Speaker at Marist College Pi Mu Epsilon Honor Society Induction
2017–2020	NYUrWIS Girls Mentor
Jan 2019	Judge for JMM undergraduate poster session
Apr 2018	Judge at The Scientista Symposium

Journal Referee Service

PAIN, PLoS Computational Biology, Physical Review E, Cognitive Neurodynamics, PLoS ONE, SIAM Journal on Applied Dynamical Systems.

Professional Development

Spring 2024	ELP Fellow
2022–2023	MAA Project NExT Fellow
Summer 2022	WRP Anti-Racist Pedagogy Fellow
Fall 2021	National Center for Faculty Development and Diversity Faculty Success Program
Aug 2021	CTLR Annual Teaching and Writing Retreat
Jan 2021	CTLR Contemporary Teaching in the Liberal Arts Series: Pandemic Teaching
Jul 2020	DLINQ Camp Design Online